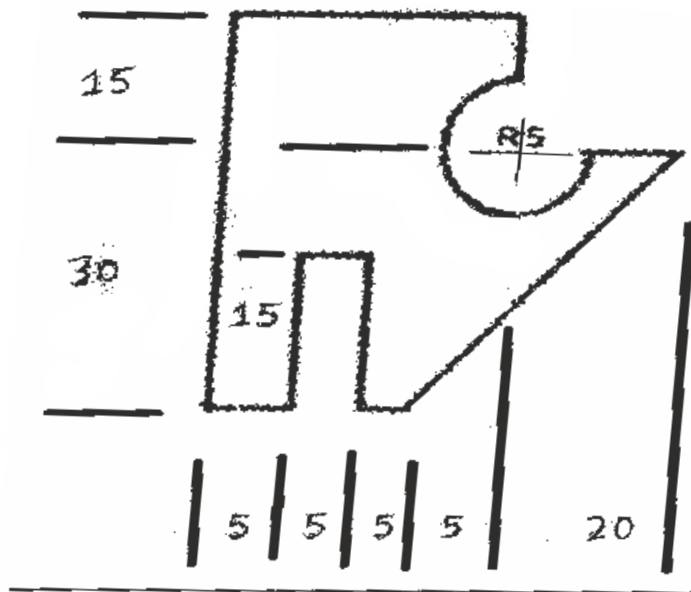


Q.25 Design and draw a die to produce sheet metal component as shown in fig 'A' of thickness of 2 mm. The component is made of Brass having an ultimate shear strength of 350N/mm^2 . Your design should include.:

1. At least two views of the die
2. Economic strip layout
3. Bill of materials
4. Press tonnage calculations



- Dimension units may be taken in mm.
- Assume suitable dimensions if not specified or clearly visible in the figure

No. of Printed Pages : 4
Roll No.

221831/181832/121832
/031832

**3rd Sem / Mechanical (Tool & Die Design),
CNC, CAD/CAM**

Subject : Press Tool-Design and Drawing

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 In blanking process, the clearance is provided on
- a) Die
 - b) Punch
 - c) Half on both die and punch
 - d) Depends on the designer choice
- Q.2 In bending operation, the metal takes the shape of
- a) Die
 - b) Punch
 - c) Half on both die and punch
 - d) Depends on the sheet thickness
- Q.3 Function of pilot is
- a) Position the punch
 - b) Position the stock strip
 - c) Position the die
 - d) All of the above

(120)

(4) 221831/181832/121832
/031832

(1) 221831/181832/121832
/031832

- Q.4 Cutting force in sheet metal operations can be reduced by
- Grinding cutting edge to be sharp
 - Hardening the tool
 - Increasing shear on tool
 - Hardening of die
- Q.5 How many dowel pins are used in press tool assembly
- 2
 - 4
 - 6
 - 1
- Q.6 For piercing operation shear is not provided on
- Stripper Plate
 - Punch
 - Die
 - Both a and b

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define embossing
- Q.8 Define Shaving
- Q.9 Define curling
- Q.10 What is strip layout?
- Q.11 Give the basic function of stripper plate.
- Q.12 Define trimming.

(2) 221831/181832/121832
/031832

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Describe with sketch the construction and working of a compound die considering a suitable example.
- Q.14 What is function of Bolster plate. Explain with a neat sketch.
- Q.15 What is shearing force?
- Q.16 Explain coining and strip guiding.
- Q.17 Enlist the factors considered while selecting press for a particular application.
- Q.18 Name any two materials that are used in making dies.
- Q.19 Write short note on die life.
- Q.20 Explain the function of pillars with the help of diagram.
- Q.21 Calculate the force required to cut a blank of 50 mm Die. from 500 mm dia and 3 mm thick sheet having shear strength of 300N/mm².
- Q.22 Write short note on shut height of press and die.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Sketch & design a progressive die to make a steel washer 60 mm outside diameter with 30 mm hole inside from 3 mm thick metallic sheet. The ultimate shear strength of the material is 320 N/mm².
- Q.24
- Differentiate between Compound die and progressive die
 - Differentiate between trimming tool and shaving tool

(3) 221831/181832/121832
/031832